

Equation Discovery for Extrapolation: Systematic Skeleton Search with a Projected Polynomial Background

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1 Introduction

A central promise of machine learning is prediction *beyond* the observed data, and standard regressors are remarkably bad at it: a polynomial diverges at a rate dictated by its degree, a random forest predicts a constant outside the convex hull of the training inputs, and a multi-layer perceptron (MLP) extrapolates linearly. All three interpolate the data in Figure 1 almost perfectly, yet fail immediately outside the training domain. Symbolic regression (SR) offers a way out: if the data-generating process *is* a closed-form formula f , recovering f up to symbolic equivalence yields exact extrapolation by construction. The price is a hard combinatorial search over expression space [Dong and Zhong, 2025], and when no formula in that language fits the data, SR has no advantage at all (Section 4).

Problem statement. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be an unknown formula over a known operator alphabet. We observe N_{train} samples (x_i, y_i) , where $x_i \sim \text{Unif}([-a, a]^d)$ and $y_i = f(x_i) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. The goal is a model $\hat{f}$ minimizing the error on test points sampled *strictly outside* the training domain, $x \sim$

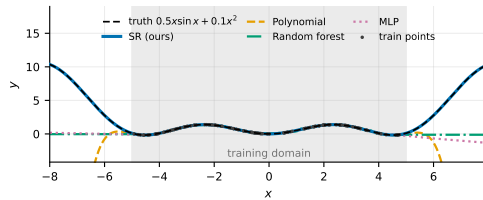


Figure 1: Extrapolation on $f(x) = 0.5x \sin x + 0.1x^2$ (128 noise-free training points on $[-5, 5]$). The baselines fit the training domain but fail outside it; ours recovers the formula exactly and coincides with the dashed truth.

$\text{Unif}([-b, b]^d \setminus [-a, a]^d), b > a:$

$$\text{MSE}_{\text{test}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (\hat{f}(x_i) - f(x_i))^2. \quad (1)$$

We contribute (i) an SR algorithm that makes the unbiased search of Kahlmeyer et al. [2025] systematic and adds two components aimed at extrapolation, namely a polynomial background solved in closed form by variable projection and cross-validation over *extrapolation horizons* for model selection, and (ii) experiments on dice-rolled formulas mapping out where the method beats polynomial, random-forest and MLP baselines and where it breaks.

2 Approach

Related work. SR systems search by genetic programming [Cranmer, 2023], gradient descent through operator layers [Kim et al., 2021, Tsoi et al., 2025], or continuous relaxation [Scholl et al., 2025], all biasing the search [Dong and Zhong, 2025] in ways that are hard to check [Kahlmeyer et al., 2025]. We adopt the assumption-free but randomized stance of Kahlmeyer et al. [2025] and make it *systematic*.

2.1 Model assumptions

We search over *skeletons*: expression trees whose internal nodes carry operators from $\mathcal{U} = \{\sin, \cos, \exp, \log, \sqrt{\cdot}, (\cdot)^2\}$ and $\mathcal{B} = \{+, -, \times, \div\}$ and whose leaves are input variables x_j or constant placeholders c ; a skeleton u with p placeholders defines a parametric family u_θ , $\theta \in \mathbb{R}^p$. Following Kahlmeyer et al. [2025], the only structural assumption on the search space is that it contains *small* expressions: the *core complexity* $k(u)$ is the number of operator nodes, and the search proceeds in order of increasing k . Operators $\log |t|$ and $\sqrt{|t|}$ are *protected*; division is not, so skeletons may have poles and invalid evaluations are penalized during fitting.

A skeleton becomes a regressor through a linear *polynomial background*,

$$\hat{f}(x) = \alpha u_\theta(x) + \sum_{j=1}^m \beta_j m_j(x), \quad (2)$$

with fixed monomials $\{m_j\} = \{1\} \cup \{x_j^q : q \leq \deg\}$ (plus $x_0 x_1$ for $d = 2$) and $\deg \in \{2, 3\}$, fixed by cross-validation in Section 3. Generalizing linear scaling [Keijzer, 2003] from $\{1\}$ to a monomial basis, the background is the crucial ingredient: the core $x \sin x$ explains Figure 1 up to a polynomial trend, so with (2) it is the best of the 69 skeletons at $k = 2$ at machine precision, while without it it ranks 24th and is pruned. The *total complexity* is $C = k(u) + \#\{j \geq 2 : \beta_j \neq 0\}$, core operators plus active background terms.

2.2 Parameter learning

For a fixed skeleton the parameters split into a nonlinear θ and a linear (α, β) , which we exploit by *variable projection*: for given θ the optimal linear coefficients are the least-squares solution

$$(\alpha^*, \beta^*)(\theta) = \arg \min_{\alpha, \beta} \|y - \alpha u_\theta(X) - M\beta\|_2^2, \quad M_{ij} = m_j(x_i), \quad (3)$$

computed in closed form. Substituting back leaves a nonlinear least-squares problem in θ only, $\min_\theta \|r(\theta)\|_2^2$ with $r(\theta) = y - \alpha^*(\theta) u_\theta(X) - M\beta^*(\theta)$, solved by a trust-region solver (finite-difference Jacobians) from the starts $\theta^{(0)} = \mathbf{1}$ and one draw from $\mathcal{N}(0, 2^2 I)$ (two for $p \geq 3$). Non-finite predictions are mapped to a large constant and skeletons with $p > 6$ placeholders discarded. Selected models are *sparsified*: terms with $|\beta_j| \text{std}(m_j) < 10^{-4} \text{std}(y)$ are dropped and the linear part refitted.

2.3 Search: exhaustive enumeration, then a diverse beam

Semantically equivalent skeletons waste budget, so trees are mapped to a canonical representative: constant subtrees fold into a placeholder ($\sin(c) \rightarrow c$), associative chains are flattened, sorted and allowed at most one constant term, parameter-absorbing compositions are rejected ($t-c$, t/c , e^{t+c} , $\log |ct|$, $(ct)^2$, $t \cdot t$, t/t , $\dots$), and aliases of the absolute value are kept only as $\sqrt{t^2}$. For $d = 1$ this leaves 1, 6, 69, 890, 12 769 skeletons with $k = 0, \dots, 4$. Such rules only remove duplicates they can *prove*, but many more are redundant with respect to (2): since α and the intercept are free, cores agreeing up to an affine transform of their output are interchangeable, and with the full background so are cores differing by a background polynomial. An *evaluation fingerprint* detects this: each skeleton is evaluated on 24 probe points for 3 generic parameter draws, the monomial basis is projected out (*background*) or not (*affine*), and the normalized result is hashed; a skeleton whose fingerprint was seen is skipped without fitting. Being heuristic, the filter is validated in Section 3.

Search runs in two phases (Algorithm 1). Phase 1 fits *every* canonical skeleton up to k_{ex} , an unbiased search in the sense of Kahlmeyer et al. [2025] that places the method in the *deterministic* branch of the taxonomy of Dong and Zhong [2025], whose trade-off – global optimality at exponentially growing cost – is the behaviour we quantify in Section 3.1. Phase 2 grows skeletons one operator at a time with a beam, expanding a parent into all canonical trees obtained by replacing a leaf with a depth-one subtree or combining any node with a fresh leaf or unary wrapper. Since ranking parents by training loss alone makes the beam greedy, parents are chosen *round-robin across root-operator groups* in order of loss, plus 5 random survivors (ε -greedy); constants are fitted on a 64-point subsample during the search and refitted afterwards. The output is a *Pareto front* $\mathcal{F} = \{\hat{f}_k\}$: the best model per core complexity $k \leq k_{\text{max}}$, kept only where the training loss strictly improves.

Algorithm 1 Complexity-ordered skeleton search; both loops stop once $\min \mathcal{F} < 10^{-10} \text{Var}(y)$.

Require: (X, y) , subsample (X_s, y_s) , operators $(\mathcal{U}, \mathcal{B})$, limits $k_{\text{ex}}, k_{\text{max}}$, width B

- 1: $\mathcal{F} \leftarrow \emptyset$; $L_{-1} \leftarrow \emptyset$ $\triangleright \mathcal{F}$: best model per core complexity
- 2: **for** $k = 0, \dots, k_{\text{max}}$ **do**
- 3: **if** $k \leq k_{\text{ex}}$ **then** $L_k \leftarrow$ all canonical skeletons with k operator nodes $\triangleright$ Phase 1
- 4: **else** $P \leftarrow$ round-robin top- B of L_{k-1} by loss across root operators $\cup$ 5 random;
- 5: $L_k \leftarrow$ canonical one-operator expansions of P (deduplicated) $\triangleright$ Phase 2
- 6: **end if**
- 7: $\text{FIT}(u, X_s, y_s) \forall u \in L_k$; update $\mathcal{F}$
- 8: **end for**
- 9: refit $\mathcal{F}$ on all training points; sparsify; **return** Pareto-optimal $\mathcal{F}$
- 10: **procedure** $\text{FIT}(u, X, y)$ $\triangleright$ variable projection, Section 2.2
- 11: $\theta^* \leftarrow \arg \min_{\theta} \|r(\theta)\|_2^2$ (multi-start trust-region NLS), $r(\theta) = y - P(\theta)P(\theta)^+y$, $P(\theta) = [u_{\theta}(X) \mid M]$
- 12: **return** θ^* , $(\alpha, \beta) = P(\theta^*)^+y$, loss $\|r(\theta^*)\|_2^2/N$
- 13: **end procedure**

2.4 Model selection by extrapolation-shaped CV

The front $\mathcal{F}$ ranges from underfit to overfit models, and picking by training loss would always return the most complex one. Because our target is (1) rather than interpolation error, we validate the way we will be tested: training points are ranked by their sup-norm distance from the domain center, and for each quantile $q \in \{0.5, 0.6, 0.7, 0.8\}$ a fold refits each candidate’s constants on the inner q fraction and validates on the outer ring, giving cross-validation over *extrapolation horizons*. With $\text{cv}_{k,f}$ the validation MSE of $\hat{f}_k$ on fold $f \in F$, candidates are scored by $s_k = \frac{1}{|F|} \sum_f \log \text{cv}_{k,f}$ and the winner is chosen by the one-standard-error rule [Hastie et al., 2009, §7.10.1]: among all k with $s_k \leq s_{k^\dagger} + \widehat{\text{se}}(s_{k^\dagger})$, $k^\dagger$ the score minimizer, the one with smallest C_k is selected and refitted on all training data. This is robust on training-loss plateaus, where CV differences are mostly noise. Baseline hyperparameters [Hastie et al., 2009, Ch. 3, 11, 15] are tuned on the *same* folds, so the comparison is fair.

Limits of the model. The hypothesis space is bounded by k_{max} (7 core operators for $d = 1$) and by the beam: formulas whose subterms only pay off jointly (e.g. $e^{-c|x|} \cos(cx)$, $k = 7$) can be missed though representable. Protected log and $\sqrt{\cdot}$ extend past their natural domains [Keijzer, 2003], unprotected division admits poles that make fitting ill-conditioned, and constant fitting is non-convex, where multi-starts mitigate but do not eliminate local minima (sin frequencies are the classic failure). The background finally biases the method towards “scaled core + polynomial” structure, from which $\sin(x_0 x_1) \log |x_0|$ gains nothing.

3 Experiments

Setup. Ground truths are sampled by a dice roll: a random tree with a prescribed number of operator nodes over one of three alphabets ($\mathcal{O}_{\text{poly}} = \{+, -, \times, (\cdot)^2\}$, $\mathcal{O}_{\text{trig}} = \mathcal{O}_{\text{poly}} \cup \{\sin, \cos\}$, and $\mathcal{O}_{\text{full}}$ with all ten operators), with leaves x_j (probability 0.7) or constants from $\pm \text{Unif}([0.2, 3])$. On a dense grid over $[-8, 8]^d$ a formula is rejected if it is invalid, unbounded ($\max |f| > 10^4$), degenerate ($\text{std}(f) < 10^{-3}$), independent of a variable, or affine despite ≥ 3 dice operators. Since dice-rolled trees often simplify ($x(x - x - x)^2 = x^3$), analyses bin by the *effective complexity* k_{eff} , the operator count of the `sympy`-simplified formula. Datasets use $N_{\text{train}} = 128$, $N_{\text{test}} = 256$, $a = 5$, $b = 8$ and $\sigma = \sigma_{\text{rel}} \text{std}(f(X))$. We report the normalized test error $\text{NMSE} = \text{MSE}_{\text{test}} / \text{Var}(y_{\text{test}})$, aggregated by medians since single blow-ups dominate means ($\text{NMSE} = 1 \approx$ predicting a constant), and the *recovery rate* [Kahlmeyer et al., 2025]: the fraction of runs matching the truth on that grid with relative RMSE below 10^{-6} . Baselines are polynomial ridge regression (degree ≤ 8), random forest and an MLP, all tuned on the folds of Section 2.4; the two commands in `README.md` reproduce all (seeded) numbers.

Search configuration. Search hyperparameters are fixed by cross-validation on the folds of Section 2.4, never on test data, using a one-factor-at-a-time grid over 48 hard datasets (Appendix A). Two factors show a consistent trend: deeper enumeration pays off, but only together with fingerprint deduplication (70.8% recovery versus 66.7% at 55% of the compute), and the background is decisive, with recovery non-decreasing in its degree and plain linear scaling ten times slower; a wider beam ($B=40$) buys almost nothing at double the cost. Since no configuration dominates, the final method runs a *portfolio* of the two best, ($k_{\text{ex}}=4$, dedup) and (deg=3), keeping the model with the better cross-validated score. This reaches 79.2%, the oracle rate of the full grid at a ninth of its cost, because an exactly recovered formula drives its cross-validated error to numerical zero, a signal no wrong configuration can imitate.

3.1 Extrapolation error and recovery

The median $\text{NMSE}_{\text{test}}$ of our method is at machine precision for *every* alphabet (Figure 2, top), since in most runs the formula is recovered exactly and exact recovery implies exact extrapolation: 85% of the 300 ground truths are recovered (100/82/73% for $\mathcal{O}_{\text{poly}}/\mathcal{O}_{\text{trig}}/\mathcal{O}_{\text{full}}$), 88% of runs reach $\text{NMSE} < 10^{-3}$ and 90% reach $\text{NMSE} < 0.1$. The baselines confirm the failure modes from Section 1: the random forest *never* reaches $\text{NMSE} < 10^{-3}$, the MLP succeeds in 1% of runs and the polynomial in 51% – almost exactly the fraction of truths that are themselves polynomials of degree ≤ 8 ; elsewhere it diverges. The aggregate hides the method’s most characteristic property: error and runtime are *bimodal*. The 75th percentile of $\text{NMSE}_{\text{test}}$ is at machine precision while the 90th is 0.13, and the median runtime of 0.03s contrasts with a worst case of 4.6 min on a

Setting	SR (ours)	Polynomial	Random forest	MLP	Recovery
d=1, poly	$8.2 \cdot 10^{-31}$	$6.6 \cdot 10^{-12}$	2.19	0.36	100%
d=1, trig	$7.1 \cdot 10^{-31}$	1.00	1.45	1.75	82%
d=1, full	$1.8 \cdot 10^{-30}$	1.07	1.25	0.96	73%
d=2, trig	$1.5 \cdot 10^{-31}$	0.25	0.74	0.15	89%
d=2, full	$1.2 \cdot 10^{-31}$	0.99	0.62	0.089	89%

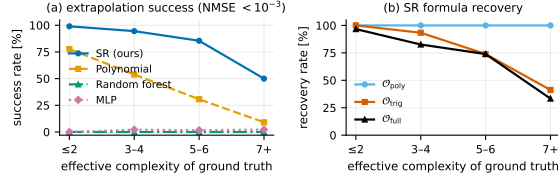


Figure 2: Main experiment. *Top*: median $\text{NMSE}_{\text{test}}$ (10^{-30} is exact recovery, 1 is as bad as predicting a constant) and SR recovery rate, from 25 ($d=1$) resp. 15 ($d=2$) noise-free formulas per alphabet–complexity cell. *Bottom* ($d=1$, 300 formulas, resolved by the effective complexity k_{eff}): (a) runs with $\text{NMSE}_{\text{test}} < 10^{-3}$, (b) runs with exact recovery.

fruitless full-depth search – either the law is recovered or it is missed, and the portfolio pays its $2.3\times$ overhead mostly where it fails anyway.

The lower panels resolve this by ground-truth complexity: recovery decays from 99% at $k_{\text{eff}} \leq 2$ over 93% and 81% to 41% at $k_{\text{eff}} \geq 7$. This is the *complexity wall* of the bounded, partially greedy search, precisely the exponential-cost trade-off that Dong and Zhong [2025] attribute to this method family; it is steeper for alphabets with log, exp and division, which create poles and near-degenerate constants. For $d=2$ recovery is 89% overall but drops from 100% at $k_{\text{eff}} \leq 4$ to 69% beyond, since the larger leaf set inflates every enumeration level; the baselines look less catastrophic there (MLP median NMSE 0.09) because most test-ring points are extreme in only *one* coordinate, so that task is partly near-interpolation.

3.2 Noise and the selection criterion

Under label noise (Appendix B), *exact* recovery drops to zero for any $\sigma > 0$, since the fitted constants inherit an $O(\sigma)$ error, so the interesting quantity is the test error. Our median NMSE degrades gracefully from 0 to 0.017 at $\sigma_{\text{rel}} = 0.2$, while every baseline sits between 0.3 and 6.6. Near-perfect extrapolations ($\text{NMSE} < 10^{-3}$) fall from 76% to 12% over that range, but usable ones ($\text{NMSE} < 0.1$) only from 92% to 56%: noise degrades the *precision* of the recovered law much faster than its *shape*.

An ablation of the selection step alone (Appendix C; all strategies pick from the *same* front) separates two effects at 10% noise. *Parsimony* is large and unambiguous: cross-validated selection returns a strictly simpler model than training-MSE selection on 92 of 120 datasets and never a more complex one

(median $C = 6$ versus 9, $p \approx 6 \cdot 10^{-17}$), though its effect on accuracy is far more modest (median error 0.025 versus 0.045, $p = 0.043$). The *fold geometry* has no measurable effect (23 wins, 24 losses, 73 ties, $p = 0.63$): both geometries pick the *identical* model in 61% of cases, because the one-standard-error rule collapses the decision onto the complexity tie-break long before fold differences matter. We keep them as the conceptually right validation at no cost; what earns its keep is parsimony.

3.3 Analysis of learned models

Recovery is judged *functionally*, and the search often returns an equivalent but different form: $\sin(2x) - \sin(2x + 2.4)$ is found as $3.73 \cos(x) \sin(x - 0.37) + 0.68$, and for the cubic $1.61 + 1.23(x - x^2)(1.08 - x)$ the core carries only x^3 and the background the rest, the division of labour intended by (2). The 30/300 runs with $\text{NMSE} > 0.1$ show three patterns: *poles* ($1/(x - 1.32 + \cos x)$), *nested non-polynomial arguments* ($\log(x - \log(x^2/1.46) - 2.05)$, reachable only through intermediates that fit poorly alone), and *frequency fitting* ($\sin(x + 2.83x^2)$, where the right skeleton is found but the nonconvex constant fit misses the frequency). Selection then falls back to a simple, poorly fitting but *bounded* model, a consequence of validating on extrapolation horizons.

Cross-validation has one blind spot, since extrapolation-shaped folds still only extrapolate to the edge of the data: the worst dataset selected $\log(e^{-x} - e^{e^x})$, unremarkable on the training box but $\text{NMSE} = 5 \cdot 10^{11}$ outside it. A *growth guard* dropping candidates that explode on the box dilated by 1.5 closes it, lowering the study’s worst case to 153 and leaving the other 299 selections bit-identical. Under noise the danger is the opposite: above the elbow the front runs flat, decorating the correct core with negligible terms (e.g. $+0.0088/\sqrt{|x|}$) that no fit-based criterion rejects but that destroy *symbolic* recovery. The one-standard-error rule picks the elbow, returning $0.504x \sin x + 0.102x^2$ at 5% noise.

4 Discussion

We built a symbolic regressor around three ideas: an unbiased search [Kahlmeyer et al., 2025] made systematic by complexity-ordered enumeration over canonical skeletons, a polynomial background projected out in closed form that makes partial explanations visible early, and model selection validating the way the model will be used. It recovers 85% of dice-rolled truths exactly and beats the baselines by orders of magnitude; their failure is structural and no tuning fixes it. Not every component earned its keep equally: the background and parsimony are decisive, the extrapolation-shaped folds *not* measurably better than interior ones, and deep enumeration only pushed the complexity wall back. More fundamentally, SR pays off only if the data-generating process *is* a formula over the chosen alphabet; otherwise a black box is preferable. Natural extensions: variable augmentation [Kahlmeyer et al., 2025], core-term deletion tests, and neural guidance [Scholl et al., 2025] of the beam.

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A Search configuration grid

Table 1: Search configurations on 48 *hard* datasets ($d=1$, $\mathcal{O}_{\text{trig}}/\mathcal{O}_{\text{full}}$, 4–8 dice operators), varied one factor at a time around the default; total time is a configuration’s cost over all 48 datasets. The last two rows arbitrate between all eleven configurations *per dataset*, by the cross-validated score s_k of Section 2.4 resp. by the smallest test error (oracle, an upper bound), and report the selected run only. Recovery differences between the levels of a factor never exceed four of the 48 datasets, so single rows indicate trends rather than resolved effects. Discussed in Section 3.

Configuration	med. NMSE _{test}	Recovery	med. #fits	med. time [s]	total time [s]
default ($k_{\text{ex}}=3$, no dedup, $B=15$, deg 2)	$2.4 \cdot 10^{-30}$	67%	1783	5.2	5152
k_ex=2	$4.6 \cdot 10^{-30}$	62%	766	6.9	9601
k_ex=4	$5.0 \cdot 10^{-30}$	62%	2929	7.4	3182
dedup=affine	$3.1 \cdot 10^{-30}$	65%	1334	4.4	4952
dedup=background	$4.8 \cdot 10^{-30}$	69%	1280	6.3	5381
k_ex=4+dedup	$8.6 \cdot 10^{-29}$	71%	1986	12.7	2838
beam=5	$9.4 \cdot 10^{-22}$	62%	1535	5.7	2942
beam=40	$2.4 \cdot 10^{-30}$	69%	2486	16.1	10797
poly_deg=0	$8.0 \cdot 10^{-29}$	62%	3282	53.8	5681
poly_deg=1	$3.1 \cdot 10^{-30}$	62%	2127	9.8	4792
poly_deg=3	$2.6 \cdot 10^{-30}$	71%	1364	6.8	3804
CV-selected per dataset	$2.4 \cdot 10^{-30}$	79%	656	2.6	3467
oracle (best test error)	$1.2 \cdot 10^{-30}$	79%	—	—	—

B Noise experiment

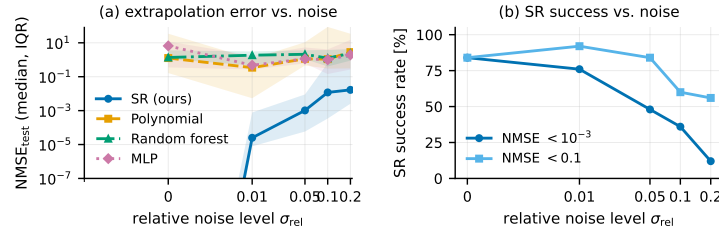


Figure 3: Noise experiment ($d=1$, $\mathcal{O}_{\text{full}}$, 25 formulas per level, $\sigma = \sigma_{\text{rel}} \text{std}(f(X))$). (a) Median NMSE_{test} (values below the axis are exact). (b) SR success rates at two thresholds. Discussed in Section 3.2.

C Model-selection ablation

Table 2: Selection ablation ($d=1$, $\mathcal{O}_{\text{full}}$, $k \in \{4, 6\}$ dice operators, 120 formulas per noise level): the same Pareto fronts, selected by extrapolation-shaped CV (ours), interior 3-fold CV, or raw training MSE. Differences are assessed by paired Wilcoxon signed-rank tests on log NMSE resp. C over the 120 shared datasets. Discussed in Section 3.2.

σ_{rel}	Selection	med. NMSE	p90 NMSE	Pr[NMSE > 1]	med. C
0	extrap-CV	$2.8 \cdot 10^{-30}$	1.14	0.11	4
0	interior-CV	$2.8 \cdot 10^{-30}$	1.21	0.12	4
0	train-MSE	$2.8 \cdot 10^{-30}$	1.76	0.12	4
0.1	extrap-CV	0.025	31.48	0.27	6
0.1	interior-CV	0.023	28.86	0.26	5
0.1	train-MSE	0.045	31.34	0.31	9